Africa Blue Carbon Initiative

Mangrove Restoration, Climate Resilience & Biodiversity Conservation

150,000 hectares • 7 African nations • USD 1 billion in climate finance

Led by the African Union Commission in partnership with Green United Development (GUD)

The Opportunity

Africa's mangrove belt holds some of the planet's highest carbon density (800 Mg C/ha), yet has degraded at alarming pace. The Africa Blue Carbon Initiative, led by the African Union Commission and Green United Development, locks away massive carbon volumes, delivers coastal protection to vulnerable communities, restores critical fisheries, and opens major new revenue streams through emerging blue carbon markets.

Carbon Density	800 Mg C/ha	5–10x terrestrial forests
Wave Protection	13–66%	per 100m mangrove belt
Fisheries Dependency	80%	West African commercial fish

Why Mangroves. Why Now.

Emerging market demand: Compliance demand for high-integrity blue carbon credits is accelerating under Article 6.2 of the Paris Agreement and CORSIA. Early bilateral transactions command premium pricing (USD 20–35/tCO2e vs. USD 10–15 VCM baseline).

Climate-resilient protection: Every 100m belt of mangroves reduces wave energy by 13–66%. A 1km-wide buffer cuts storm surge depths by 5–50cm. As climate impacts accelerate, intact mangrove systems are among the most cost-effective natural defenses available.

Fisheries restoration: Up to 80% of commercial fish species in West Africa depend on mangrove nurseries. Restoration directly strengthens regional fisheries productivity and supports 1.5+ million coastal residents.

Geographic Footprint: 7 Target Nations

Country	Restoration Target	Annual Value	Context
Nigeria	30,000 ha	USD 40.5M	Largest mangrove cover
Cameroon	18,000 ha	USD 32.6M	Sixth-largest in Africa
Mozambique	35,000 ha	USD 3.6M	Second-largest extent
Madagascar	20,000 ha	USD 3.8M	2% of global mangroves
Guinea	15,000 ha	USD 2.6M	Ramsar wetlands
Tanzania	20,000 ha	USD 1.0M	Largest East African
Benin	12,000 ha	USD 0.6M	High per-hectare value
TOTAL	**150,000 ha**	**USD 84.6M/yr**	

Financial Architecture

Metric	Value
Phase I (2026–2031)	USD 453 million
Phase II (2031–2036)	USD 547 million
Total Investment	USD 1 billion
Gross annual credit potential	~7.05 million tCO2e/year
Net annual issuable credits (20% buffer)	~5.64 million tCO2e/year
Reference carbon price (conservative)	USD 15/tCO2e
Indicative annual revenue	~USD 84.6 million/year
30-year gross revenue potential	~USD 2.54 billion

Pricing reflects conservative voluntary market assumptions. Compliance or Article 6.2 pricing would uplift value.

By 2036: Integrated Outcomes

Climate & Carbon: 10M tCO2e/year sequestration verifiable for carbon markets. Permanence anchored in sediment carbon (1,500–3,245 Mg C/ha). Supporting the 1.5°C goal.

Biodiversity & Fisheries: 1,000+ species habitat regenerated. 80% of West African fish species benefit from mangrove nursery recovery. Commercial fisheries productivity restored.

Community & Livelihoods: 1.5+ million coastal dwellers with improved livelihoods. Women & youth in sustainable value chains. Food security strengthened.

Policy & Finance: Blue carbon integrated into NDCs and NAPs. Five+ blue bonds or carbon mechanisms with fair benefit-sharing.

Methodology: Community-Based Ecological Restoration

The initiative prioritises **CBEMR**—restoring hydrological conditions for natural regeneration. Emphasises native species diversity, natural zonation, coordinated seed provenance mapping, rigorous MRV (geospatial and participatory), and conservative carbon crediting (Gold Standard, Verra, Plan Vivo). Green United Development brings expertise in community-centered carbon projects, remote sensing monitoring (ODIN technology), and large-scale restoration implementation across the continent.

Permanence & Risk Management

Sediment carbon permanence: 1,500–3,245 Mg C/ha at depth (multi-century storage).

Non-permanence buffer: 20% conservative buffer per international standards.

Landscape-scale design: Large contiguous blocks reduce displacement and enable co-management.

Climate-informed siting: High restorable potential in climate-resilient zones.

Governance: Annual third-party audits and transparent reporting.

Strategic Alignment

Supports **Agenda 2063** (inclusive growth and blue economy), **2050 Africa Maritime Strategy** (marine conservation), **Paris Agreement Articles 5 & 6** (nature-based solutions), **Kunming–Montreal Biodiversity Framework** (ecosystem restoration), **AU Climate Strategy** (resilient blue economies), and **National NDCs** across all seven member states.

Implementation Model

Implementing agencies: African Union Commission (lead), in collaboration with Green United Development, Regional Economic Communities, and national environment, fisheries, and biodiversity authorities.

GUD role: Community-centered carbon project development, ODIN remote sensing monitoring and verification, partnership mobilization, and large-scale restoration delivery.

Financing: Blended finance (public, private, philanthropic)—blue bonds, debt-for-nature swaps, catalytic grants.

Technical: Leading institutions for restoration, MRV, carbon accounting, biodiversity monitoring.

Governance: Community stewardship integrating traditional knowledge and co-management. Gender and youth emphasis.

Next Steps

• Detailed technical and financial due diligence with GUD and financing partners

• Site-level baseline assessments and hydrological studies

• Community engagement and benefit-sharing negotiations

• Capacity building and institutional arrangements

• Finalisation of financing structures and carbon market pathways

For more information:
African Union Commission, Directorate of Sustainable Environment & Blue Economy
Green United Development (GUD)
February 2026